Theoretical Performance of Airborne Coherent Doppler Lidar for Turbulence Detection Using Spectral Width Estimation of Heterodyne-Detected Signal

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***Abstract*— The theoretical performance of an airborne coherent Doppler lidar (CDL) for turbulence detections is shown. Irregular wind fluctuation, which causes aircraft vertical acceleration, is defined as turbulence. The detection method is based on spectral width estimation of a heterodyne-detected signal. The equations for simulation are derived by referring to the one for a Doppler radar. Performance theory on a CDL is additionally utilized to derive the equations. Probabilities of missed and false hazard indications are simulated with an example set of input parameters. The simulated turbulence detectable range roughly agrees with past experimental results. The contents of this article can contribute to the future establishment of minimum operational performance standards (MOPSs) for turbulence detection using an airborne CDL.**

***Index Terms*— Airborne, coherent Doppler, turbulence, wind.**

I. Introduction

TURBULENT atmosphere is the main critical factor in aircraft operation at both the cruising and landing-approach altitudes. Toward safer aircraft operation, the Radio Technical Commission for Aeronautics (RTCA) has published the document (DO)-220 [1]. This document has stated minimum operational performance standards (MOPSs) for low-level wind shear (LLWS) detections using an airborne Doppler radar. Only rainy conditions have been assumed in this document, but clear sky has not been considered. On the other hand, a coherent Doppler lidar (CDL) [2], [3] is suitable for the detection of the abovementioned turbulent phenomena under clear sky conditions. Therefore, many flight test campaigns using airborne CDLs have been reported [4], [5], [6], [7], [8], [9], [10], [11], [12]. In particular, LLWS detections at low altitudes have been demonstrated successfully [5], [11]. This led to our recent work for the performance simulation theory of this application based on DO-220 [13], toward future establishment of MOPS for CDLs.

Irregular wind fluctuation that causes aircraft vertical acceleration has been defined as the word "turbulence" in [14]. Regarding turbulence detection, the past feasibility study [15] stated that the performance of existing CDLs (e.g., a CDL that was developed in the Safeavio project of Japan Aerospace Exploration Agency (JAXA) [11], [12]) could not satisfy the needs for practical use because of very low aerosol backscattering at cruising altitudes. However, there have been many demonstrations on turbulence detections with airborne CDLs. For example, turbulence detection was demonstrated using a 2-μm airborne CDL [6]. Furthermore, turbulence detections using 1.5-μm airborne CDLs have been reported in several cases [7], [8], [9], [10], [11], [12].

Here, we show the theoretical performance of an airborne CDL for turbulence detection. The simulation scenario is close to the scenario of our recently presented paper [13], but the method of turbulence detection uses spectral width estimation of a heterodyne-detected signal instead of line-of-sight (LOS) wind velocity estimation. The turbulence detection theory using spectral width estimation is based on the theory for Doppler radars [14], which was referred in the revised version of DO-220 (i.e., DO-220A [16]). As mentioned above, the contents of this article have a strong relationship with the related literature [13], [14], [16]. The theories of a CDL [17], [18] are utilized to adjust to the theory in [14].

Direct detection lidars (DDLs) [19], [20], [21] are outside the scope of this article because of the following reason. DDLs utilize intensity fluctuation of a detected signal for turbulence detections. On the other hand, the referred literature [14] has been based on spectral width estimation of a detected signal. The analogy in turbulence detection is much stronger for CDLs than for DDLs.

In the following, the turbulence detection method is explained in Section II. The simulation scenario is shown in Section III. The equations for performance simulation of turbulence detection are shown in Section IV. The input and output parameters are summarized in Section V, and also, analogy to the related literature [13], [14], [16] is summarized. In Section VI, an example of the simulation of turbulence

Manuscript received 9 March 2023; revised 18 May 2023; accepted 9 July 2023. Date of publication 12 July 2023; date of current version 27 July 2023. *(Corresponding author: Shumpei Kameyama.)*

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Digital Object Identifier 10.1109/TGRS.2023.3294516

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detection performance is shown using a practical CDL specification.

II. Turbulence Detection Method

The configuration of an airborne CDL for turbulence detection is basically the same as the one for LLWS detection [13, Fig. 1] and is not shown here. It is a pulsed system, and wind sensing is performed by: 1) range gating of a heterodyne-detected signal and 2) spectral analysis of each gated signal using the fast Fourier transform (FFT). There are some differences in the signal processing between LLWS and turbulence detections. The LLWS detection is performed by utilizing a change of LOS wind velocity for adjacent ranges [13]. On the other hand, turbulence detection is performed based on spectral width estimation of a detected signal [14] by utilizing spectral broadening caused by irregular wind change. While the spectral broadening of a Doppler radar is dominated by turbulence, the broadening of a CDL is also influenced by some instrumental parameters. A wind velocity standard deviation (WVSD, m/s) within a range gate along with the LOS is estimated by compensating these instrumental parameters. An estimated WVSD is given by

$$\Delta v_e = \sqrt{w_e^2 - \left(\frac{\lambda}{2}\cdot\frac{\sqrt{2\ln 2}}{2\pi\Delta t}\right)^2 - \left(\frac{\lambda}{2}\cdot\frac{\Delta f}{2\sqrt{2\ln 2}}\right)^2} \quad (1)$$

where λ is the wavelength (m), Δf is the linewidth of a laser source (FWHM, Hz), and Δt is the pulsewidth (FWHM, s). The WVSD relates to the energy dissipation rate in the case of Kolmogorov turbulence [22, eq. (2)]. It should be noted that the term of optical frequency chirping caused by pulsed operation has been neglected. This equation can be derived from [13, eq. (5)]. w_e in (1) is the estimated spectral width of a heterodyne-detected signal (m/s) at a certain range and is given by

$$w_e = \frac{\lambda}{2T_G}\sqrt{\frac{\sum_{i=b_L}^{b_U}\left[(i-p)^2\cdot S(i)\right]}{\sum_{i=b_L}^{b_U} S(i)}} \quad (2)$$

where i is the frequency (or velocity) bin number and T_G is the rage gate width (s). $S(i)$ is the spectral intensity, which is obtained by FFT and spectral accumulation (after noise-floor rejection), p is the peak frequency bin number of $S(i)$. b_U and b_L are the upper and lower boundaries where $S(i)$ is continuously larger than the noise level around the spectral peak. If the signal level is low, $S(i)$ may be larger than noise level only at $i = p$. In such a case, $p = b_U = b_L$ and zero estimation occurs regarding spectral width (i.e., $w_e = 0$). The abovementioned two cases regarding spectral width estimation are shown schematically in Fig. 1. Even if the estimated spectral width is not zero, it can be very small, and the square root of (1) becomes imaginary (not real) in some cases. In such cases, an indication of turbulence detection is not alerted.

The root-mean-square (rms) vertical load excursion (g) is the critical factor for aircraft operation. This includes the influence of WVSD and is estimated as [14]

$$\sigma_{\Delta n}^E = \frac{Q\Delta v_e}{U} \quad (3)$$

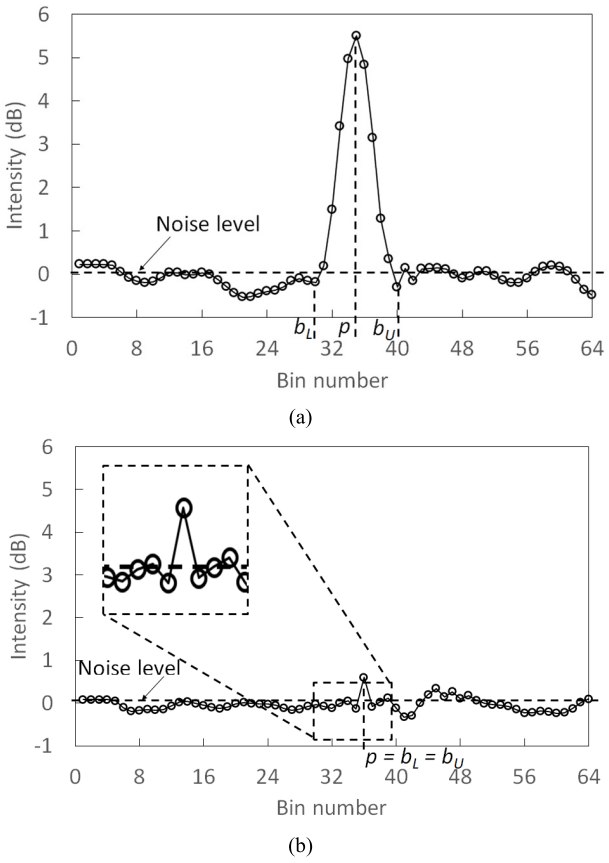


Fig. 1. Schematic of spectrum in cases of (a) high SNR and (b) low SNR (zero-estimation case). Lower and upper boundaries for spectral width estimation are additionally shown. Intensity has been normalized by a noise level.

where Q is the aircraft scale factor [g/(m/s)]. This parameter relates WVSD to rms vertical load excursion, and it depends on aircraft type, flight altitude, and airspeed [14]. U is the volume compensation factor. This parameter relates wind fluctuation in a probe volume of a CDL to in situ wind fluctuation at a location of an aircraft. In the case of CDLs, this is approximately expressed as [14]

$$U \cong 1 - \frac{55}{27}\cdot\frac{1}{1.339\pi}\cdot\left(\frac{1.339L}{\sigma_r}\right)^5\int_0^\infty \frac{x^{\frac{3}{2}}e^{-x}}{\left[1+\left(\frac{1.339L}{\sigma_r}\right)^2 x\right]^{\frac{17}{6}}}\cdot M'\left[2,\frac{5}{2},x\right]dx \quad (4)$$

where L is the turbulence scale length (m). σ_r in (4) is the radial second moments of Gaussian beam illumination function (m), which are given by [14]

$$\sigma_r = \frac{\Delta t}{2\sqrt{2\ln 2}}. \quad (5)$$

M' in (4) is calculated using the Gamma function with variables α, γ, and ζ as [14]

$$M'(\alpha,\gamma,\zeta) = \frac{\Gamma(\gamma)}{\Gamma(\alpha)}\sum_{n=0}^{\infty}\frac{\Gamma(n+\alpha)\zeta^n}{\Gamma(n+\gamma)n!}. \quad (6)$$

The derivation process of the volume compensation factor [i.e., (4)] for a Doppler radar has been shown [14]. For general CDLs, the condition of $R\sigma_\theta \ll \sigma_r$ (σ_θ: transverse second

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moment of Gaussian beam illumination function) is satisfied since a laser beam is approximately pencil-like. Equation (4) can be derived from [14, eq. (I-8)] under this condition. It should be noted that Δv_e in (3) has been replaced by spectral width in the case of Doppler radar [14] since spectral broadening is dominated by turbulence. The estimated rms vertical excursion, which is obtained by (3), is compared with a threshold value. A detection alert is triggered when the estimated rms vertical load excursion ($\sigma^E_{\Delta n}$) is larger than the threshold value. The threshold has been set based on the category regarding turbulence strength [14]

$$\begin{aligned}
&\sigma^E_{\Delta n} \le 0.1\text{g Light turbulence}\\
&0.1\text{g} < \sigma^E_{\Delta n} \le 0.2\text{g Moderate turbulence}\\
&0.2\text{g} < \sigma^E_{\Delta n} \le 0.3\text{g Moderate to Severe turbulence}\\
&0.3\text{g} < \sigma^E_{\Delta n} \le 0.6\text{g Severe turbulence}\\
&0.7\text{g} < \sigma^E_{\Delta n} \text{ Extreme turbulence.}
\end{aligned}$$

III. Simulation Scenario

The simulation scenario for turbulence detection is schematically shown in Fig. 2. As the first step, simulation for range dependence of wideband signal-to-noise ratio [SNR, in the followings, SNR(R), R: range (m)] is performed using atmospheric and CDL parameters. In the next step, the range-dependent performance of spectral width estimation is simulated using SNR, signal processing parameters, in addition to atmospheric and aircraft parameters which impact turbulence conditions. Then, a turbulence detection performance is simulated using the performance of spectral width estimation, signal processing parameters, and aircraft parameters. Equations for the simulation are shown in Section IV.

IV. Equations for Simulation

Equations for SNR calculation are exactly the same as those in our recently presented article [13, eqs. (2)–(4)] and are not shown here.

Regarding the estimation performance of spectral width, a Monte Carlo simulation is used to derive the relation between SNR and the performance of spectral width estimation. The simulation procedure of the Monte Carlo simulation is based on the procedure of [13, Fig. 4]. The signal and noise are randomly generated in the spectral domain with Gaussian and white shapes depending on the SNR [18]. The true value of spectral width for a generated signal is given by [13]

$$w_T = \sqrt{\left(\frac{\lambda}{2}\cdot\frac{\sqrt{2\ln 2}}{2\pi\,\Delta t}\right)^2 + \Delta v_T^2 + \left(\frac{\lambda}{2}\cdot\frac{\Delta f}{2\sqrt{2\ln 2}}\right)^2} \tag{7}$$

where Δv_T is the true value of WVSD, which is given by [14]

$$\Delta v_T = \frac{U}{Q_m}\cdot\sigma^T_{\Delta n} \tag{8}$$

where $\sigma^T_{\Delta n}$ is the true value of rms vertical load excursion (g) and Q_m is the mean of aircraft scale factor [g/(m/s)]. The probability density function of the aircraft scale factor is given by [14]

$$\text{pdf}_Q(Q) = \frac{1}{(2\pi)^{1/2} g_Q}\exp\left[-\frac{(Q-Q_m)^2}{2g_Q^2}\right] \tag{9}$$

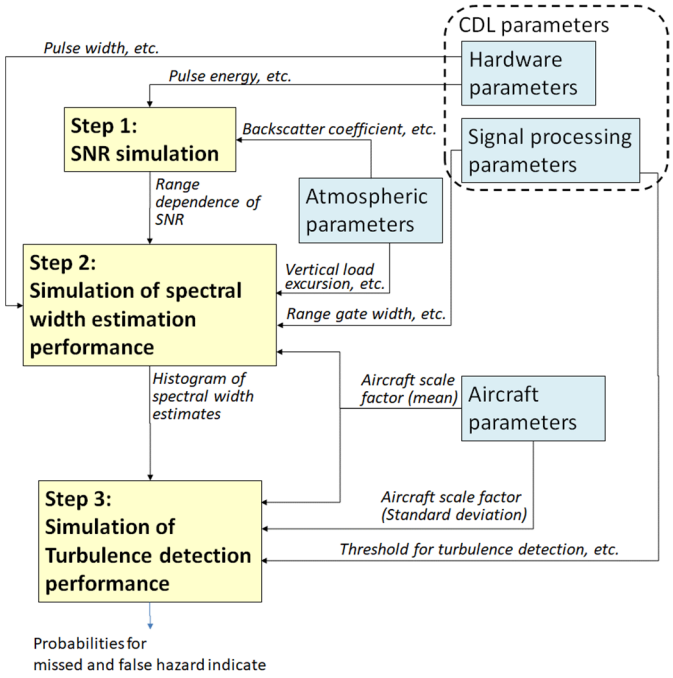


Fig. 2. Schematic of simulation scenario.

where g_Q is the standard deviation of aircraft scale factor [g/(m/s)]. It should be noted that the aircraft scale factor of this article is a statistical parameter with the concept of "one-size-fits-all" [14]. In an alternative concept of "strapped" [14], this parameter becomes deterministic and depends on each aircraft type (e.g., ranging from Boeing A320 to B747). On the other hand, the concept of "one-size-fits-all" depends only on rough aircraft category (classes A, B, and C [14]). Therefore, this concept is employed here for more general simulation theory with accepting error to some extent.

Estimations of wind velocity and spectral width are performed on the simulated spectrum after incoherent spectral accumulation. The abovementioned spectral width estimation is performed iteratively. After that, the relation between SNR and histogram of estimates is derived by changing the SNR.

An example histogram of spectral width estimates at a certain SNR is shown in Fig. 3. There are empirically two components, Gaussian and zero-estimation ones. The zero estimation occurs when the peak intensity of spectrum is low [see Fig. 1(b)]. Therefore, zero estimation basically means low data quality and low SNR. The histogram of spectral width estimates can be approximated with a least-square fitting using the following empirical function:

$$\begin{aligned}
&\text{PDF}_W\left(w_e, \text{SNR}(R), \sigma^T_{\Delta_n}\right)\\
&= p\left(\text{SNR}(R), \sigma^T_{\Delta_n}\right)\cdot\delta(0)\\
&\quad + \frac{\left(1-p\left(\text{SNR}(R), \sigma^T_{\Delta_n}\right)\right)}{(2\pi)^{\frac{1}{2}} g_W\left(\text{SNR}(R), \sigma^T_{\Delta_n}\right)}\\
&\quad\cdot\exp\left[-\frac{\left(w_e - w_m\left(\text{SNR}(R), \sigma^T_{\Delta_n}\right)\right)^2}{2g_W^2\left(\text{SNR}(R), \sigma^T_{\Delta_n}\right)}\right]
\end{aligned} \tag{10}$$

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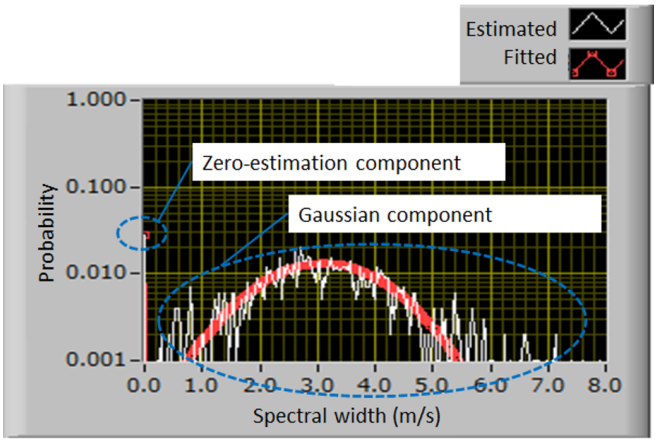


Fig. 3. Schematic of histogram of spectral width estimates at a certain SNR with fit functions.

where $w_m(\mathrm{SNR}(R), \sigma^T_{\Delta_n})$ is the mean of Gaussian component (m/s), $g_W(\mathrm{SNR}(R), \sigma^T_{\Delta_n})$ is the standard deviation of Gaussian component (m/s), and $p(\mathrm{SNR}(R), \sigma^T_{\Delta_n})$ is the zero-estimation probability. These are the fitting parameters.

The relation between SNR and estimation performance of spectral width is shown in Fig. 4 with additional fitting functions. The fitting functions for probabilities of the zero-estimation $[P_{\mathrm{zero}}(R)]$ and Gaussian components [(standard deviation $(\sigma_w(R))$ and mean $(\sigma_m(R))$] are given as follows:

$$P_{\mathrm{zero}}(R) = f' \cdot \exp\left[-\frac{(\mathrm{WSNR} - e')^2}{g'}\right] \tag{11}$$

$$\sigma_w(R) = \begin{cases} h' + (i' - h') \cdot \exp\left[-\dfrac{(\mathrm{WSNR} - o')^2}{j'}\right], \\ \qquad\qquad \text{if } \mathrm{WSNR} \le o' \\ i' \cdot \exp\left[-\dfrac{(\mathrm{WSNR} - o')}{k'}\right], \quad \text{else} \end{cases} \tag{12}$$

$$\sigma_m(R) = m' - (m' - l') \cdot \exp\left[-\frac{(\mathrm{WSNR} - e')}{n'}\right] \tag{13}$$

where WSNR is the wideband SNR (in the unit of dB). f', g', h', i', j', k', l', m', and n' are the fitting parameters. o' is the SNR (in the unit of dB) at which inflection appears in the fitting function [see Fig. 4(b)]. e' is the SNR (in unit of dB) at a signal detection probability of 0.6. Regarding the performance curve of (11)–(13), the SNR is equivalently improved approximately proportional to $\sqrt{N}$ if $N > 10$ (N: spectral accumulation number, see [13, eq. (7)] [22]. The performance calculation is also needed for wind velocity estimation to derive the parameter e'. The schematic on how to derive this parameter can be known from Figs. 5 and 6, and [13, eqs. (8)–(10)].

The reason for using wideband SNR in Fig. 4 is explained here. Detectability [23] is the ratio between intensity of peak frequency bin and standard deviation of noise level, and this can be an alternative candidate to express the SNR. However, detectability includes the influence of spectral broadening

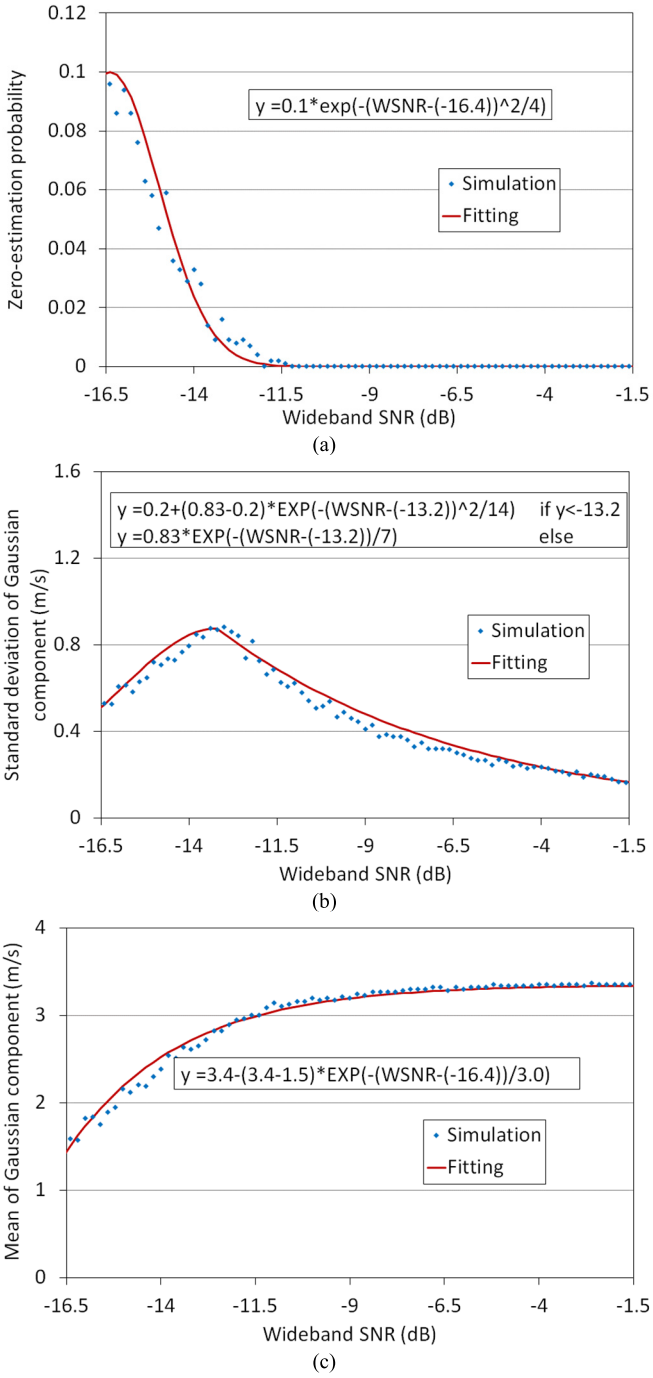


Fig. 4. Example of relation between wideband SNR. (a) Zero-estimation probability. (b) Standard deviation of Gaussian component. (c) Mean of Gaussian component with fit function under the condition of laser wavelength: 1.55 μm, range gate width: 100 m, spectral width: 8.5 m/s, sampling frequency: 100 MHz (complex sample), accumulation number: 100, and velocity search range: -30 to 30 (m/s).

(and turbulence) even in the cases of the same wideband SNR. This means that the scaling of horizontal axis in Fig. 4 changes depending on turbulence conditions. This makes the understanding of Fig. 4 complex. Furthermore, the wind sensing performance depends not only on detectability but also on spectral width. Therefore, the expression using detectability does not unify the performance curve of Fig. 4 and the calculation of the performance curve must be done for each turbulence condition in either method (wideband SNR

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or detectability). Using wideband SNR can, at least, isolate the process of SNR calculation and derivation of signal detection performance. This is the reason for using wideband SNR.

The following investigations are performed regarding the performance curves of Fig. 4. The zero-estimation probability in Fig. 4(a) does not approach to the value of one (i.e., 100%) even in low SNR cases. Frequency bines in a spectrum around the peak frequency bin often become larger than the noise level due to the statics of noise. In such situations, the estimated spectral width becomes a finite value, which is larger than zero. Furthermore, in low SNR cases, the estimated spectral width and its standard deviation tend to be low since the numbers of frequency bins around the peak, which are larger than the noise level tend to be small. The inflection in Fig. 4(b) and the change of mean in Fig. 4(c) appear due to these reasons.

The following assumptions are made: 1) U is deterministic and 2) Δv, Q_m, and $\sigma_{\Delta n}^T$ are statistically random and independent of each other. Under these assumptions, the histogram regarding estimated rms vertical load excursion ($\sigma_{\Delta n}^E$) is given by [14]

$$P\left(\sigma_{\Delta n}^E, \sigma_{\Delta n}^T, \mathrm{SNR}(R)\right) = N' \int_0^\infty \frac{\mathrm{pdf}_W\left(w_e, \mathrm{SNR}(R), \sigma_{\Delta n}^T\right) \cdot \mathrm{pdf}_Q\left(\frac{U\sigma_{\Delta n}^E}{\Delta v_e}\right)}{\Delta v_e} d\Delta v_e \tag{14}$$

where N' is the correction factor to normalize the total probability as 1. In this equation, the relation between w_e and Δv_e [i.e., (1)] should be noted. The probabilities for missed and false hazard indications are calculated as [14]

$$P_M(g_H, \mathrm{SNR}(R)) = 1 - \int_{g_T}^\infty P\left(\sigma_{\Delta n}^E, g_H, \mathrm{SNR}(R)\right) d\sigma_{\Delta n}^E \tag{15}$$

$$P_{\mathrm{FD}}(g_L, \mathrm{SNR}(R)) = \int_{g_T}^\infty P\left(\sigma_{\Delta n}^E, g_L, \mathrm{SNR}(R)\right) d\sigma_{\Delta n}^E \tag{16}$$

where g_T, g_L, and g_H are the threshold, must-not indicate, and must indicate hazardous vertical load excursion (g), respectively.

V. Input and Output Parameters of Simulation

The input and output parameters of the simulation are summarized in Tables I and II. There are some parameters and symbols, which are not described in this article, but they are denoted in the related literature [13, eqs. (2)–(4) and (7)]. The class 1 aircraft at an altitude of 5000 ft (i.e., 1524 m) with an airspeed of 250 knot indicated airspeed (kias) has been considered. Atmospheric parameters correspond to the values at the altitude of 1000 m, which are the ones denoted in Appendix B of [13]. The hardware parameters are the same as in [13], and these are based on an existing airborne CDL [11], [12] with some empirically reasonable modifications. The parameters are shown in Table IV of Appendix A. The signal processing parameters, which relate to hazardous turbulence detection, are the same as in DO-220A [16].

TABLE I
Input Parameters of Simulation

Parameters		Symbol	Value	Unit
Atmosphere	Aerosol backscatter coefficient	β	4.0×10^{-8}	/m/sr
	Extinction to aerosol backscatter ratio	R_L	50	sr
	Molecular absorption coefficient	α_m	3.7×10^{-6} *1	/m
	Atmospheric refractive index structure constant	C_n^2	1.5×10^{-16}	$m^{-2/3}$
	Turbulence scale length	L	500*2	m
Aircraft	Mean of aircraft scale factor	Q_m	0.04514 *2, *3	g•s/(m/s)
	Standard deviation of aircraft scale factor	g_Q	0.004071 *3	g•s/(m/s)
CDL (Signal processing)	Threshold for hazardous vertical load excursion	g_T	0.13	g
	Must-not indicate vertical load excursion	g_L	0.1 *2, *4	g
	Must indicate vertical load excursion	g_H	0.3 *2, *4	g
	Range gate width	T_G	6.7×10^{-7}	s
	LOS wind velocity search range	v_{search}	-30 to 30	m/s
	LOS measurement rate	T	0.5*5	s

*1 U.S. standard atmosphere
*2 Corresponding spectral width: 3.4 m/s and 1.2 m/s for must and must-not indicate vertical load excursions
*3 Correspondent to class 1 aircraft at altitude of 5,000 ft with airspeed of 250 kias [14].
*4 Requirement in MOPS of DO-220A [16]
*5 Corresponding accumulation number: 500

The relation between the equation in this article and that in the related literature [13], [14], [16] is also shown in Table II. Some equations are perfectly the same as those in [14] and [16], but there are some changes for other equations because of the different characteristics between a CDL and a Doppler radar. The combination of equations of this article and [13] realizes the analogous simulation theory of turbulence detection for a CDL to the theory for a Doppler radar. [16, eqs. (C-1) and (C-2)], which are referred in Table II, are shown in Appendix B for reference. Other equations in this table are known easily by seeing this article [13] and [14].

VI. Simulation Results

A simulation is performed using the example parameters of Table I. As the first step, the relation between SNR and spectral width estimation is simulated. The obtained fitting parameters for the must and must-not indicate vertical load excursion are shown in Table III. The range dependence on SNR is shown in Fig. 5. The threshold, which corresponds to 80% detection probability at must indicate vertical load excursion, is also shown in the figure. This threshold can be derived using [13, eq. (8)]. Fig. 5 shows that practical CDL parameters in Table IV can realize a wind measurable range of 6.7 km under the condition of must indicate hazardous turbulence. The simulation for turbulence detection is performed for only the region where SNR is higher than this threshold. The range dependences on mean and standard deviation of width estimates [$\sigma_m(R)$ and $\sigma_w(R)$ in (12) and (13)] are shown in Fig. 6 for both the must and must-not indicate vertical load excursion. The zero-estimation probabilities are also plotted

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TABLE II
OUTPUT PARAMETERS OF SIMULATION AND RELATION OF EQUATIONS BETWEEN THIS ARTICLE AND [13], [14], [16]

Category	Contents of equation	Symbol	Correspondent equation			
			This paper (CDL)	Related paper [13] (CDL)	[14, 16] (Radar)	Perfectly same?
SNR	Range dependence of SNR	$SNR(R)$	-	(2), (3), (4)	(C-1) of [16]	No
Wind sensing performance	Volume compensation factor	U	(4), (6)	-	(I-8), (I-6) of [14]	Yes
	Radial second moments of Gaussian beam illumination functions	σ_r	(5)	-	(I-3) of [14]	No
	Signal spectrum width	w_T	(7)	-	(1) of [14]	No
	Wind velocity standard deviation (WVSD)	Δv_T	(8)	-	(1) of [14]	No
	Aircraft scale factor	$PDF_Q(Q)$	(9)	-	(30) of [14]	Yes
	Spectral accumulation number	N	-	(7)	-	No
	Range dependence of standard deviation of good estimates and detection probability	$PDF(v_e, SNR(R))$ $P_{rob}(R)$ $P_{rec}(R)$	-	(8), (9), (10)	(C-2) of [16]	No
	Range dependence of standard deviation of Gaussian estimates and zero-estimation probability	$PDF_W(w_e, SNR(R), \sigma_{\Delta_n}^T)$ $P_{zero}(R)$ $\sigma_w(R)$ $\sigma_m(R)$	(10), (11), (12), (13)	-	-	No
Turbulence detection performance	Histogram regarding estimated RMS vertical load excursion	$P\left(\sigma_{\Delta n}^E, \sigma_{\Delta n}^T . SNR(R)\right)$	(14)	-	(6) of [14]	No
	Probability of Missed hazard indication	$P_M(g_H, SNR(R))$	(15)	-	(33) of [14]	Yes
	Probability of false hazard indication	$P_{FD}(g_L, SNR(R))$	(16)	-	(34) of [14]	Yes

TABLE III
FITTING PARAMETERS FOR SIMULATED RESULTS

Parameter	Value	
	Must indicate	Must-not indicate
e (dB)	-22.1	-24.5
f	0.1	0.1
g	2.5	2.5
h	0.2	0.2
i	0.44	0.22
j	15	15
k	8	9
l	1.2	0.48
m	1.4	0.48
n	3	3
o (dB)	-19.9	-22.7

for the second vertical axis of the same figure. The decreased SNR at the long-range region increases the zero-estimation probability. The estimated spectral width approaches to a true value at the high SNR region, but discrepancy occurs at the low SNR (i.e., long-range) region. The stepwise plots appear in the standard deviation of spectral width estimates. This appears at the inflection of the performance curve [i.e., WSNR $= o'$, see Fig. 4(b)]. The range dependences on the probabilities regarding the missed and the false hazard indication are shown in Fig. 7. The probabilities for missed and false hazard indications become worse for longer ranges. The stepwise plots appear because of the same reason as in Fig. 6. The criterion of 0.15 both for the must indicate and must-not indicate vertical load excursion is also plotted in the figure. This criterion corresponds to the requirement in DO-220A [16]. Both the missed and false indicate probabilities are less than the abovementioned criterion for the range up to 5.6 km. The limiting factor of this case is the missed indication of must indicate vertical load excursion. This result of detectable range agrees with the result in the past turbulence detection experiments with approximately the same detection parameters [11]. Fig. 8 shows the turbulence detectable range, which is obtained by changing the detection threshold for vertical load excursion. The limiting factor is also shown in the second axis of the figure. The detectable range becomes maximum at the threshold of 0.11 g where the limiting factor changes from false to missed indication. Even though the threshold is lower than the must-not indicate value (i.e., 0.1), the detectable range can be kept to some extent. This is due to the following reason. When the estimated spectrum width is small and the square root of (1) becomes negative, the turbulence detection process is not performed (see also Section II). This means that false detection does not appear in this case and makes the probability of false detection low. This is one of the characteristics of CDLs that Doppler radars do not have. When the threshold is set too low or too large, there are no detectable ranges. This can be seen at both sides of the figure. Figs. 5–8 are just example simulation results at one altitude. However, simulations for other cases are not shown here since the most important scope of this article is to clarify the simulation theory for turbulence detection performance. Atmospheric parameters for some other altitudes are shown in [15] and Appendix B of [13].

The turbulence detection range in Fig. 8 is significantly shorter than the requirement in the DO-220A (12 nautical miles $\approx$ 22.2 km) [1], even though the assumed flight altitude is the lowest case for the class 1 aircraft [14]. The performance of wind sensing of a CDL degrades at upper altitudes because of the lower aerosol backscatter coefficient [13]. For example, the aerosol backscatter coefficient at a cruising altitude of 12 km is 2.8×10^{-10}/m/sr [13], which is less than -20 dB compared with the simulation condition of Fig. 5. In such a condition, the SNR curve in Fig. 5 becomes

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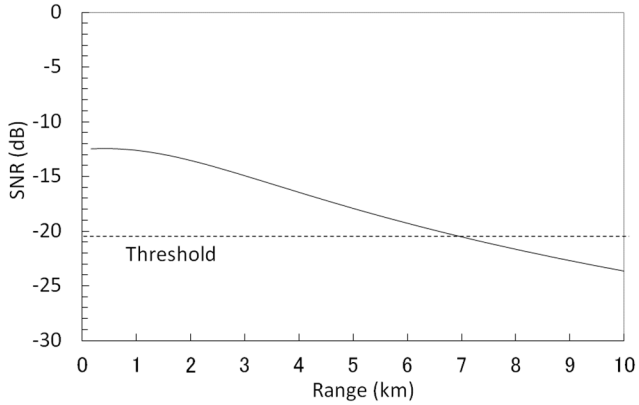


Fig. 5. Range dependence of SNR. The threshold line corresponding to 80% detection probability for must indicate vertical load excursion is also plotted.

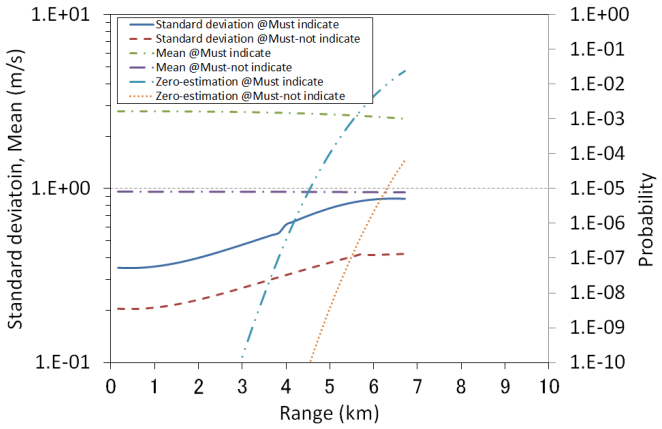


Fig. 6. Range dependences on mean and standard deviation of spectral width estimates for both the must and must-not indicate vertical load excursion.

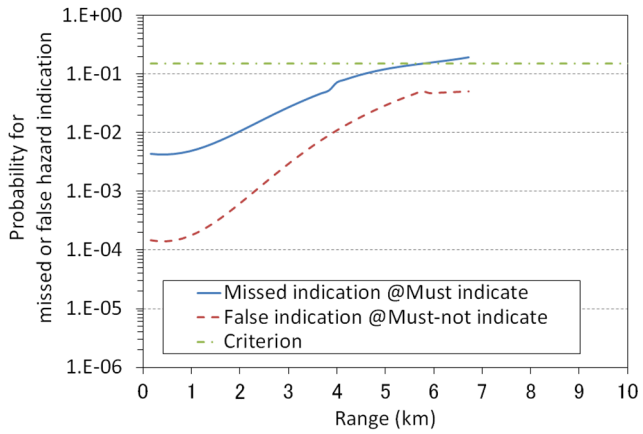


Fig. 7. Range dependences on probabilities regarding the missed and the false hazard indication. The criterion of 0.15 both for must indicate and must-not indicate vertical load excursion is also plotted.

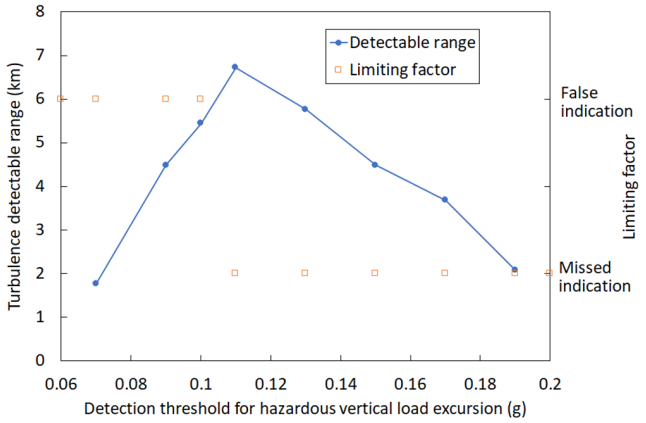


Fig. 8. Turbulence detectable range and its limiting factor versus detection threshold for hazardous vertical load excursion.

lower than the signal detection threshold for whole ranges. This fact means that the performance of the CDL of this article cannot be applicable to the turbulence detections at cruising altitudes, even though the simulation theory itself is applicable. However, the simulated results of this article imply the feasibility of short-range applications at landing-approach altitudes. For example, an advisory function for pilots with turbulence information has been developed, and the value of this function has been evaluated positively [11]. The theory and results of this article can be useful for the future establishment of MOPS for this advisory function. In addition, the existing CDL [12], which has been referred in this article (CDL in JAXA's Safeavio project), has shown the following further potential. This CDL demonstrated a wind measureble range of 1.3 km with a cumulative probability of 90% even at a cruising altitude of 12 km [12]. This performance has been realized by extending the range gate width to 2 μs (corresponding to the range resolution of 300 m) and increasing LOS measurement rate to 1 s. The abovementioned performance on measurable range implies the potential application to the gust load alleviation [24], [25], [26], [27], [28], even though more careful feasibility study is needed. The combination of the advisory function at landing-approach altitudes and the gust load alleviation at cruising altitudes can be a practical strategy [12].

VII. Conclusion

The theoretical performance of an airborne CDL for turbulence detections was shown. The method of turbulence detection is based on spectral width estimation of a heterodyne-detected signal. The simulation procedure is based on the theory for Doppler radars, which has been referred in DO-220A. The theories of a CDL are utilized to adjust to the theory of a Doppler radar.

After the explanation of simulation theory, the analogy to the theory for a Doppler radar [14] was summarized. One example of performance simulation was performed based on an existing airborne CDL. It was shown that wind sensing could be realized for a range of 6.7 km. Furthermore, it realized that a missed and false hazard indicate a probability of less than 0.15 for the must and must-not indicate vertical load excursion at a range of 5.6 km. This result agreed with the detectable range in the past turbulence detection experiments with the approximately same detection parameters. The turbulence detection range shown in this article is significantly shorter than the requirement in the DO-220A. However, the turbulence detection performance of this article implies the feasibility of short-range applications. The contents of this article can contribute to the future establishment of MOPS for turbulence detection using an airborne CDL.

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TABLE IV
CDL HARDWARE PARAMETERS

Parameters	Symbol	Value	Unit
Laser wavelength	λ	1.55×10^{-6}	m
Laser linewidth	Δf	2.5×10^{5} *1	Hz
Pulse energy	E	3×10^{-3}	J
Pulse width	Δt	5×10^{-7}	s
Pulse repetition frequency	PRF	1,000	Hz
Receiving aperture diameter	D	0.11	m
Aperture correction factor	A_C	0.71	
Beam focusing range	F_F	20,000	m
Receiving bandwidth (wideband)	B	1×10^{8}	Hz
Sampling rate (in complex sample)	-	1×10^{8}	Hz
System efficiency	η	0.1	
Far-field heterodyne-detection efficiency	η_F	0.4	

*1 This value is 2.5×10^{5} in [13] which is an unreported error.

APPENDIX A

The hardware parameters of a CDL used in the simulation are the same as [13]. The values are listed in Table IV. The symbols correspond to those in [13].

APPENDIX B

Equations (C-1) and (C-2) of [16] are shown here. Equation (C-1) of [16] has shown the range dependence of SNR. This has been given by

$$\mathrm{SNR}(R)=\frac{P_R G_R^2 T_R \theta_B \phi_B \Delta R_R \pi^6 |k_w|^2 Z_e 10^{-18}}{4R\lambda_R^2 (4\pi)^3 (kT_K F_n) L_R} \tag{17}$$

where P_R is the radar transmitted pulse power (W), G_R is the radar antenna gain, λ_R is the wavelength of radio wave (m), T_R is the radar pulsewidth (s), θ_B is the azimuth antenna beamwidth (rad), ϕ_B is the elevation antenna beamwidth (rad), ΔR_R is the radar range bin size (m), $|k_w|$ is the rain dipole moment constant (=0.92), Z_e is the reflectivity factor ($\mathrm{mm^6/m^3}$), k is Boltzmann's constant (J/K), T_K is the temperature (K), L_R is the radar system loss, and F_n is the noise figure. Equation (C-2) of [16] has shown the variance of mean velocity estimate (i.e., the square of estimated WVSD). This has been given by

$$\Delta {v_e}^2=\frac{\lambda_R^2}{4MT_S^2}\cdot\left(\frac{\sigma_{mn}}{4\sqrt{\pi}}+2\sigma_{mn}^2\cdot\frac{1}{\mathrm{SNR}(R)}+\frac{1}{12}\cdot\frac{1}{\mathrm{SNR}(R)^2}\right) \tag{18}$$

where M is the number of samples used in FFT, T_S is the time between uniform samples (i.e., pulse repetition interval, seconds), and σ_{mn} is the normalized spectral width of the radar weather signal [$=2\sigma_w T_S/\lambda_R$, σ_w: radar spectral width (m/s)].

ACKNOWLEDGMENT

The authors would like to thank Dr. Naoki Matayoshi of JAXA, Tokyo, Japan, for his meaningful advise regarding the simulation of this article.

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